

Assignment for Week 3

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Package Loading

```
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(haven)
```

Assignment

- 1) 19.2.3 (a) (p. 405)
- 2) 19.2.6 (p. 406)
- 3) 11.1.4 (c, e, f) (p. 207)
- 4) 11.2.4 (p. 213)
- 5) 11.2.6 (p. 213)
- 6) 11.3.1 (p. 217)

7) Which of the following plots of the same data is clearer? Why?



8) Use the Afghanistan dataset to answer the following:

- What if we take the average elevation of each language, which is spoken in the second highest altitudes on average?
- What language's speakers are farthest away from Kabul on average?
- (Extra Credit) What is the most geographically isolated city with a population more than 1000 in Afghanistan?

9) Correct the common mistakes in the following code snippets

a) Make the following TRUE

```
mean(c(3, 2, 1, NA)) == 2
```

b) Make the following TRUE

```
df <- data.frame(var1 = c(35, 40, 45)); length(df[1])==3
```

c) Make the following TRUE

```
df <- data.frame(var1 = c(35, 40, 45)); dplyr::filter(df, "var1"==40)==40
```

d) Make the following produce a plot of the points described in x and y.

```
x <- c(3, 10, 4); y <- c(2, 3, 4); ggplot()+geom_point(x, y)
```

e) Produce the string 'Iowa'

```
library(dplyr);
df <- data.frame(var1 = c(35, 40, 45), var2=c("Iowa", "France","Mexico"));
dfs <- df%>%filter(var2="Iowa")%>%select("var2");
as.character(dfs[[1]])
```

10) Make a plot of Afghanistan which tells us about the geography of language diversity. Avoid overplotting, where the points overlap too much. Make sure to label your x and y axis, and to make a nice title. Check out <https://cran.r-project.org/web/packages/ggthemes/vignettes/ggthemes.html> and <http://shinyapps.org/apps/RGraphCompendium/index.php> for ideas.